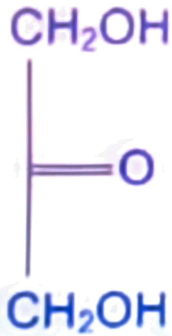
Which classification fits this sugar?

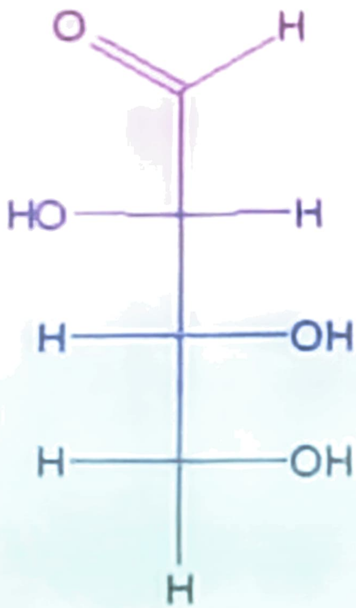


- Aldotriose
- Aldotetrose
- Aldopentose
- Aldohexose
- Ketotriose
- Ketotetose
- Ketopentose
- Ketohexose

Carbohydrates

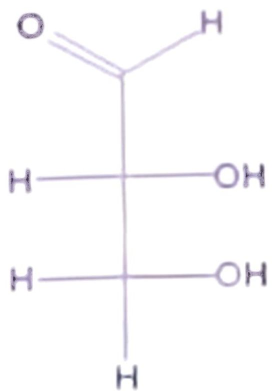
- The most abundant organic molecules in nature
- Provide a significant fraction of the energy in the diet of most organisms
- Important source of energy for cells
- Can act as a storage form of energy
- Can be structural components of many organisms
- Can be cell-membrane components mediating intercellular communication
- Can be cell-surface antigens
- Can be part of the body's extracellular ground substance
- Can be associated with proteins and lipids
- Part of RNA, DNA, and several coenzymes (NAD^+, $NADP^+$, FAD, CoA)

Which classification fits this sugar?



- Aldotriose
- Aldotetrose
- Aldopentose
- Aldohexose
- Ketotriose
- Ketotetose
- Ketopentose
- Ketohexose

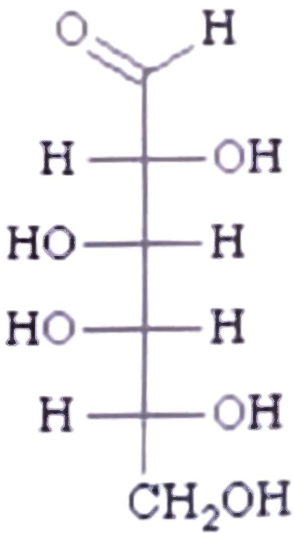
Which classification fits this sugar?



- Aldotriose
- Aldotetrose
- Aldopentose
- Aldohexose

- Ketotriose
- Ketotetose
- Ketopentose
- Ketohexose

Which classification fits this sugar?



- Aldotriose
- Aldotetrose
- Aldopentose
- Aldohexose
- Ketotriose
- Ketotetose
- Ketopentose
- Ketohexose

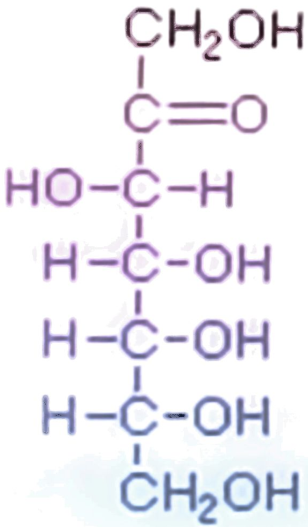
Monosaccharides

Polyhydroxy aldehydes or ketones that can't easily be further hydrolyzed

"Simple sugars"

Number of carbons	Name	Example
3	Trioses	Glyceraldehyde
4	Tetroses	Erythrose
5	Pentoses	Ribose
6	Hexoses	Glucose, Fructose
7	Heptoses	Sedoheptulose

Which classification fits this sugar?



- Aldotriose
- Aldotetrose
- Aldopentose
- Aldohexose

- Ketotriose
- Ketotetose
- Ketopentose
- Ketohexose
- Ketoheptose

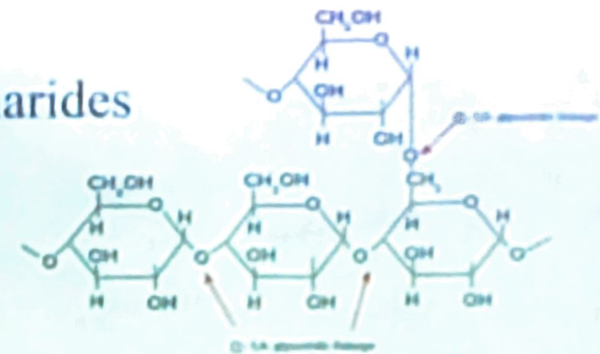
Oligosaccharides

Hydrolyzable polymers of 2-6 monosaccharides

Disaccharides composed of 2 monosaccharides
Examples: Sucrose, Lactose

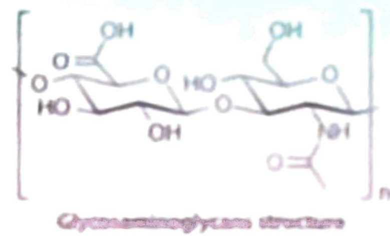
Polysaccharides

Hydrolyzable polymers of > 6 monosaccharides



Homopolysaccharides:
polymer of a single type of monosaccharide:
Examples: Glycogen, Cellulose

Heteropolysaccharides:
polymer of at least 2 types of monosaccharide:
Example: Glucosaminoglycans



Isomerism

Structural isomers

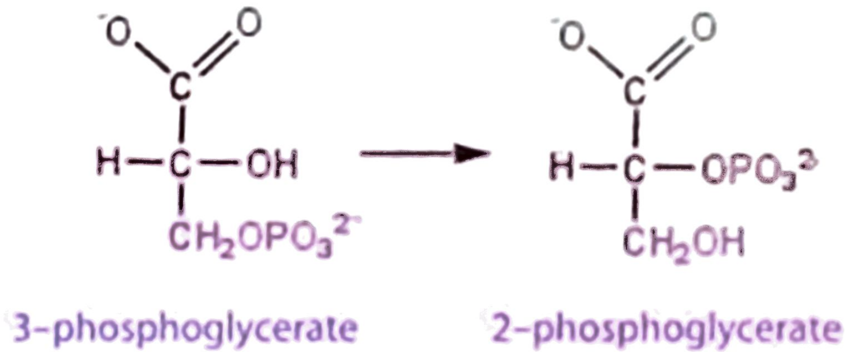
Compounds with the same molecular formula but with different structures

Functional group isomers: with different functional groups,

E.g. glyceraldehyde and dihydroxyacetone

Positional isomers: with substituent groups on different C-atoms,

E.g.



Oligosaccharides

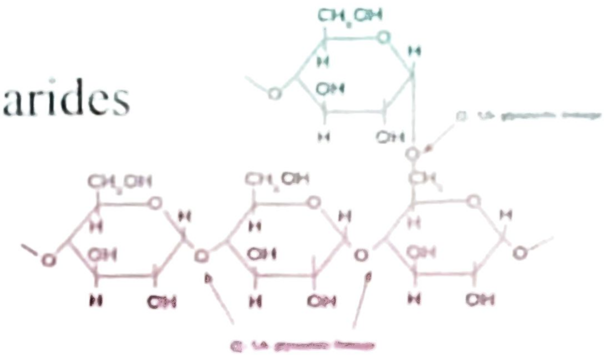
Hydrolyzable polymers of 2-6 monosaccharides

Disaccharides composed of 2 monosaccharides

Examples: Sucrose, Lactose

Polysaccharides

Hydrolyzable polymers of > 6 monosaccharides



Homopolysaccharides:

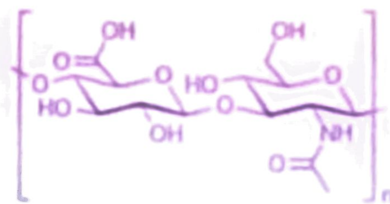
polymer of a single type of monosaccharide:

Examples: Glycogen, Cellulose

Heteropolysaccharides:

polymer of at least 2 types of monosaccharide:

Example: Glucosaminoglycans



Glycosaminoglycans structure

Isomerism

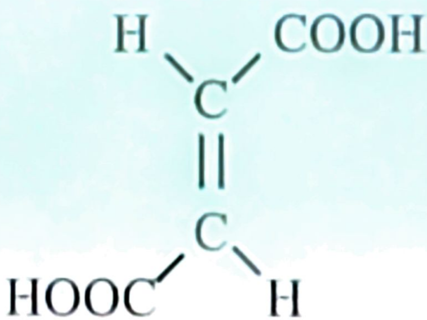
Stereoisomers

Compounds with the same molecular formula, functional groups, and position of functional groups but

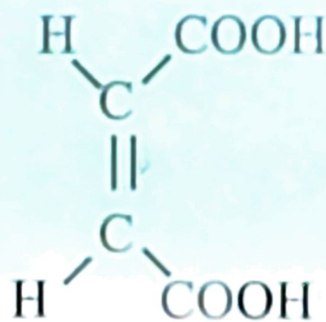
with different conformations

cis-trans isomers

with different conformation around double bonds



Fumaric acid (*trans*)



Maleic acid (*cis*)

Isomerism

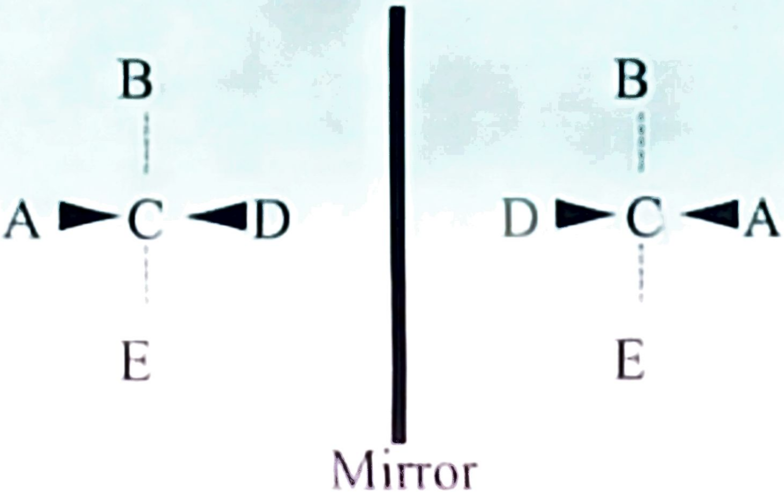
Stereoisomers

Compounds with the same molecular formula, functional groups, and position of functional groups but

with different conformations

optical isomers

with different conformation around chiral or asymmetric carbon atoms



The mirror images can't be superimposed on each other, i.e. they are different

The mirror image isomers constitute an ***enantiomeric pair***; one member of the pair is said to be the ***enantiomer*** of the other

Isomerism

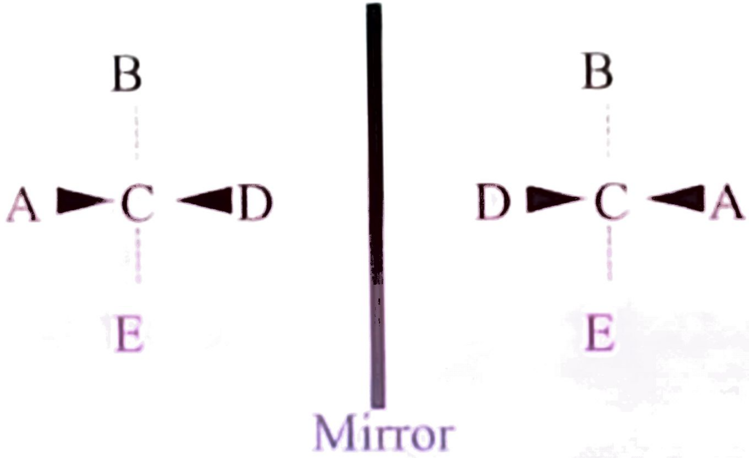
Stereoisomers

Compounds with the same molecular formula, functional groups, and position of functional groups but

with different conformations

optical isomers

with different conformation around chiral or asymmetric carbon atoms



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Isomerism

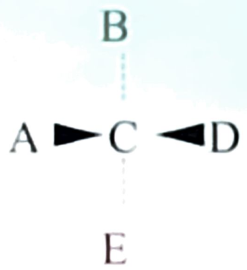
<u>Stereoisomers</u>

Compounds with the same molecular formula, functional groups, and position of functional groups but

<u>with different conformations</u>

<u>optical isomers</u>

with different conformation around chiral or asymmetric carbon atoms



The carbon C is asymmetric if A, B, D, and E are four different groups

The four different groups A, B, D, and E can be arranged in space around the C-atom in two different ways to generate two different compounds

Reference compound for optical isomers is the simplest monosaccharide with an asymmetric carbon: glyceraldehyde

O H

C ← C-atom 1

|

H- C - OH ← C-atom 2 (an asymmetric carbon)

| ← C-atom 3

CH_2OH

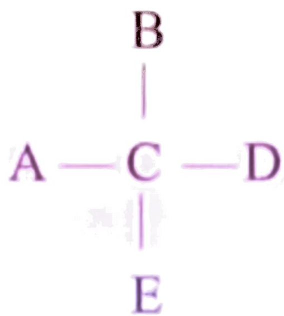
D-Glyceraldehyde	L-Glyceraldehyde
O H	O H
C	C
|	|
H- C - OH	HO- C - H
|	|
CH_2OH	CH_2OH

D-Glyceraldehyde L-Glyceraldehyde

D-Glyceraldehyde is assigned to be the isomer that has the hydroxyl group on the right when the aldehyde group is at the top in a Fischer projection formula.

It is also dextrorotatory, so it is also D(+)-Glyceraldehyde

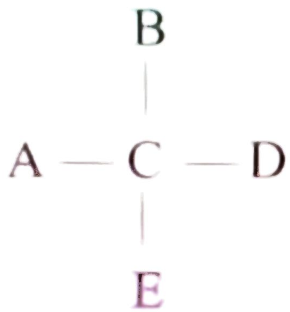
Isomerism



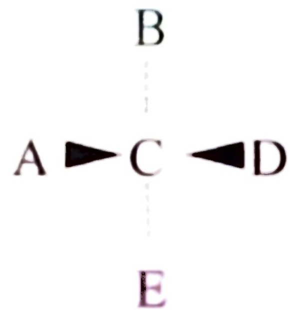
Fischer projection formula

Perspective formula

Isomerism



Fischer projection formula



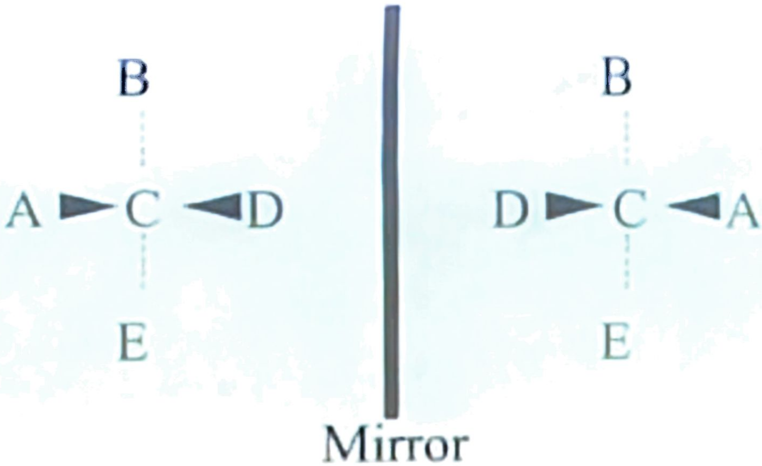
Perspective formula

If a compound has n asymmetric carbon atoms then

there are 2^n different optical isomers

Number of carbon atoms	Aldose/Ketose	Number of asymmetric carbon atoms	Number of optical isomers
3	Aldose	1	2
4	Aldose	2	4
5	Aldose	3	8
6	Aldose	4	16
3	Ketose	0	-
4	Ketose	1	2
5	Ketose	2	4
6	Ketose	3	8

Isomerism



One member of an ***enantiomeric pair*** will rotate a plane of polarized light in a clockwise direction.
It is said to be *dextrorotatory* which is labelled (+)

The other member of the pair will then rotate the light in a counter-clockwise direction.
It is said to be *levorotatory* which is labelled (-)

Isomerism

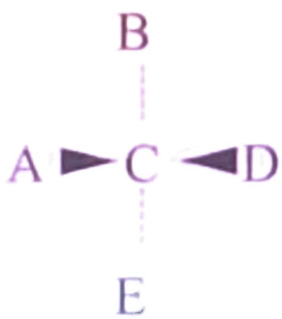
Stereoisomers

Compounds with the same molecular formula, functional groups, and position of functional groups but

with different conformations

optical isomers

with different conformation around chiral or asymmetric carbon atoms

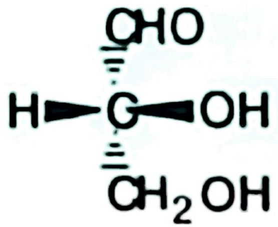


The carbon C is asymmetric if A, B, D, and E are four different groups

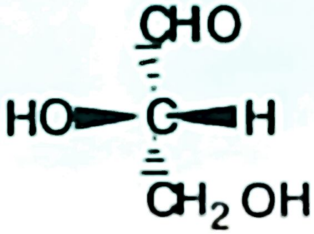
The four different groups A, B, D, and E can be arranged in space around the C-atom in two different ways to generate two different compounds

Monosaccharides

- Glyceraldehyde contains a stereocenter and exists as a pair of enantiomers.



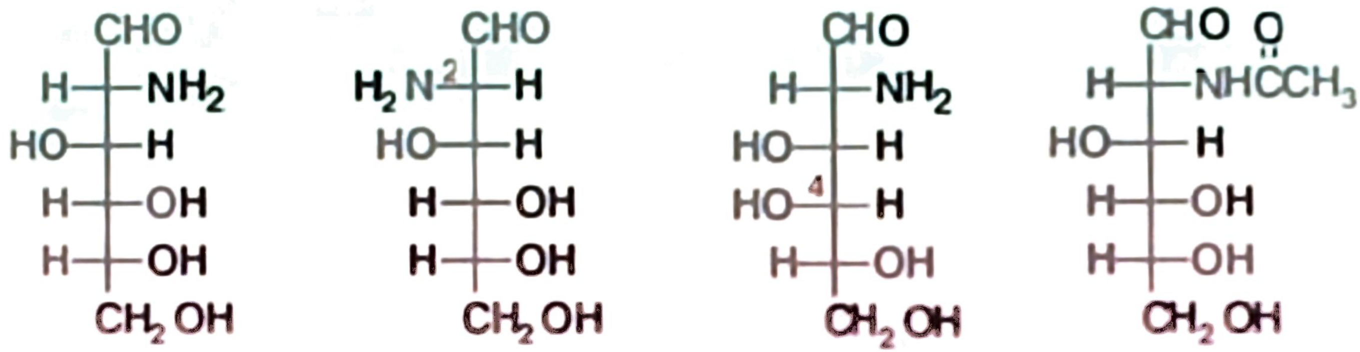
(*R*)-Glyceraldehyde



(*S*)-Glyceraldehyde

D. Amino Sugars

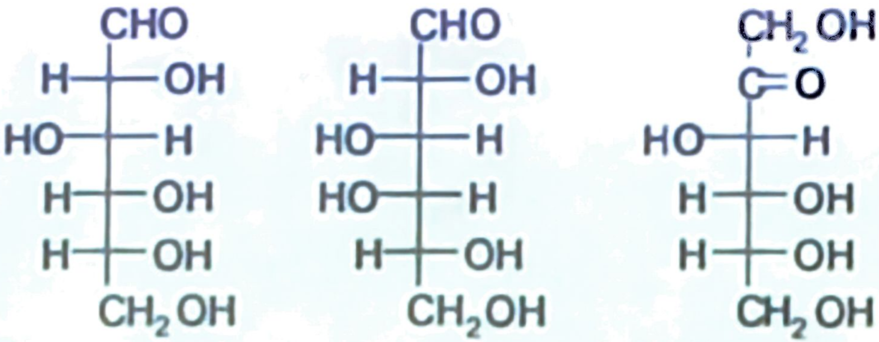
- **Amino sugar:** a sugar that contains an -NH_2 group in place of an -OH group.
  - only three amino sugars are common in nature.
  - *N*-acetyl-D-glucosamine is a derivative of D-glucosamine.



D-Glucosamine D-Mannosamine D-Galactosamine *N*-Acetyl-D-glucosamine

D,L Monosaccharides

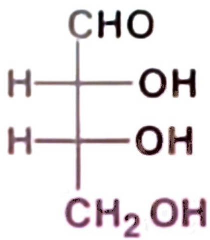
- And the three most abundant hexoses.



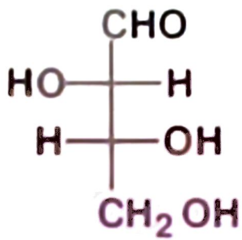
D-Glucose D-Galactose D-Fructose

D,L Monosaccharides

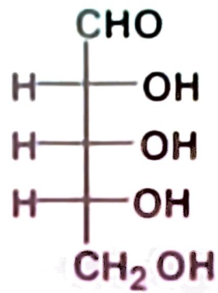
- Here are the two most abundant D-aldotetroses and the two most abundant D-aldopentoses in the biological world.



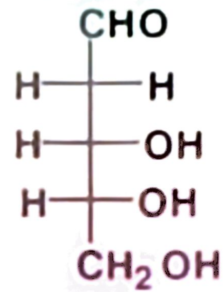
D-Erythrose



D-Threose



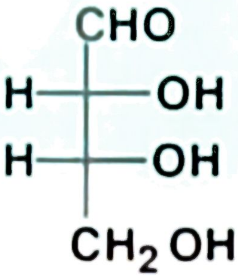
D-Ribose



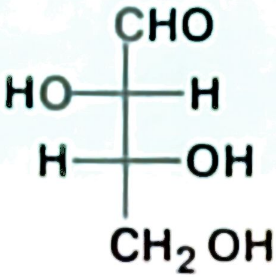
2-Deoxy-D-ribose

D,L Monosaccharides

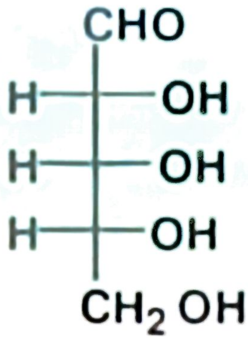
- Here are the two most abundant D-aldotetroses and the two most abundant D-aldopentoses in the biological world.



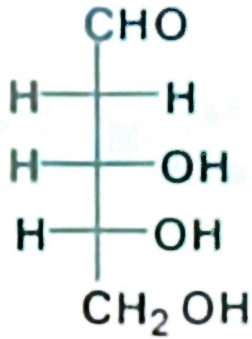
D-Erythrose



D-Threose



D-Ribose



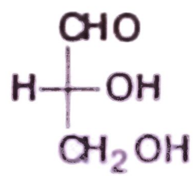
2-Deoxy-D-ribose

D,L Monosaccharides

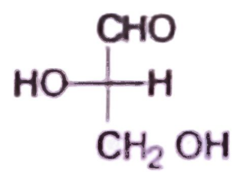
- According to the conventions proposed by Fischer.
  - D-monosaccharide: a monosaccharide that has the same configuration at its penultimate carbon as D-glyceraldehyde; that is, its -OH is on the right when written as a Fischer projection.
  - L-monosaccharide: a monosaccharide that has the same configuration at its penultimate carbon as L-glyceraldehyde; that is, its -OH is on the left when written as a Fischer projection.

C. D,L Monosaccharides

- In 1891, Emil Fischer made the arbitrary assignments of D- and L- to the enantiomers of glyceraldehyde.



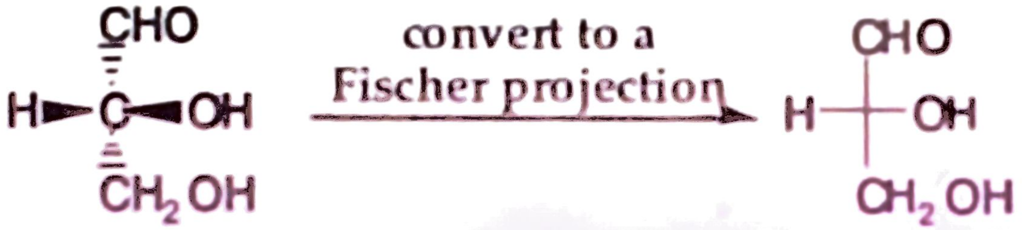
D-Glyceraldehyde



L-Glyceraldehyde

B. Fischer Projections

- **Fischer projection:** a two dimensional representation for showing the configuration of carbohydrates.
  - horizontal lines represent bonds projecting forward.
  - vertical lines represent bonds projecting to the rear.
  - the only atom in the plane of the paper is the stereocenter.



(*R*)-Glyceraldehyde (three-dimensional representation)

(*R*)-Glyceraldehyde (Fischer projection)

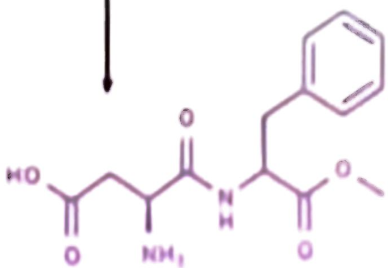
Monosaccharides

- Glyceraldehyde contains a stereocenter and exists as a pair of enantiomers.

(*R*)-Glyceraldehyde (*S*)-Glyceraldehyde

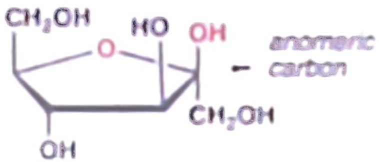
- Monosaccharides are colorless crystalline solids, very soluble in water, but only slightly soluble in ethanol.
  - sweetness relative to sucrose:

Carbohydrate	Sweetness Relative to Sucrose	Artificial Sweetener	Sweetness Relative to Sucrose
Fructose	1.74	Saccharin	450
Invert sugar	1.25	Acesulfame-K	200
Sucrose (table sugar)	1.00	Aspartame	160
Honey	0.97		
Glucose	0.74		
Maltose	0.33		
Galactose	0.32		
Lactose (milk sugar)	0.16		



25.2 Cyclic Structure

- Monosaccharides have hydroxyl and carbonyl groups in the same molecule and exist almost entirely as five- and six-membered cyclic hemiacetals.
  - **anomeric carbon**: the new stereocenter created as a result of cyclic hemiacetal formation.
  - **anomers**: carbohydrates that differ in configuration at their anomeric carbons.



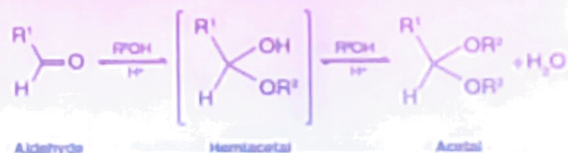
Anomer – the name given to two diastereomeric monosaccharides that are epimers at the anomeric carbon. This is C-1 in aldoses, and C-2 in the case of fructose

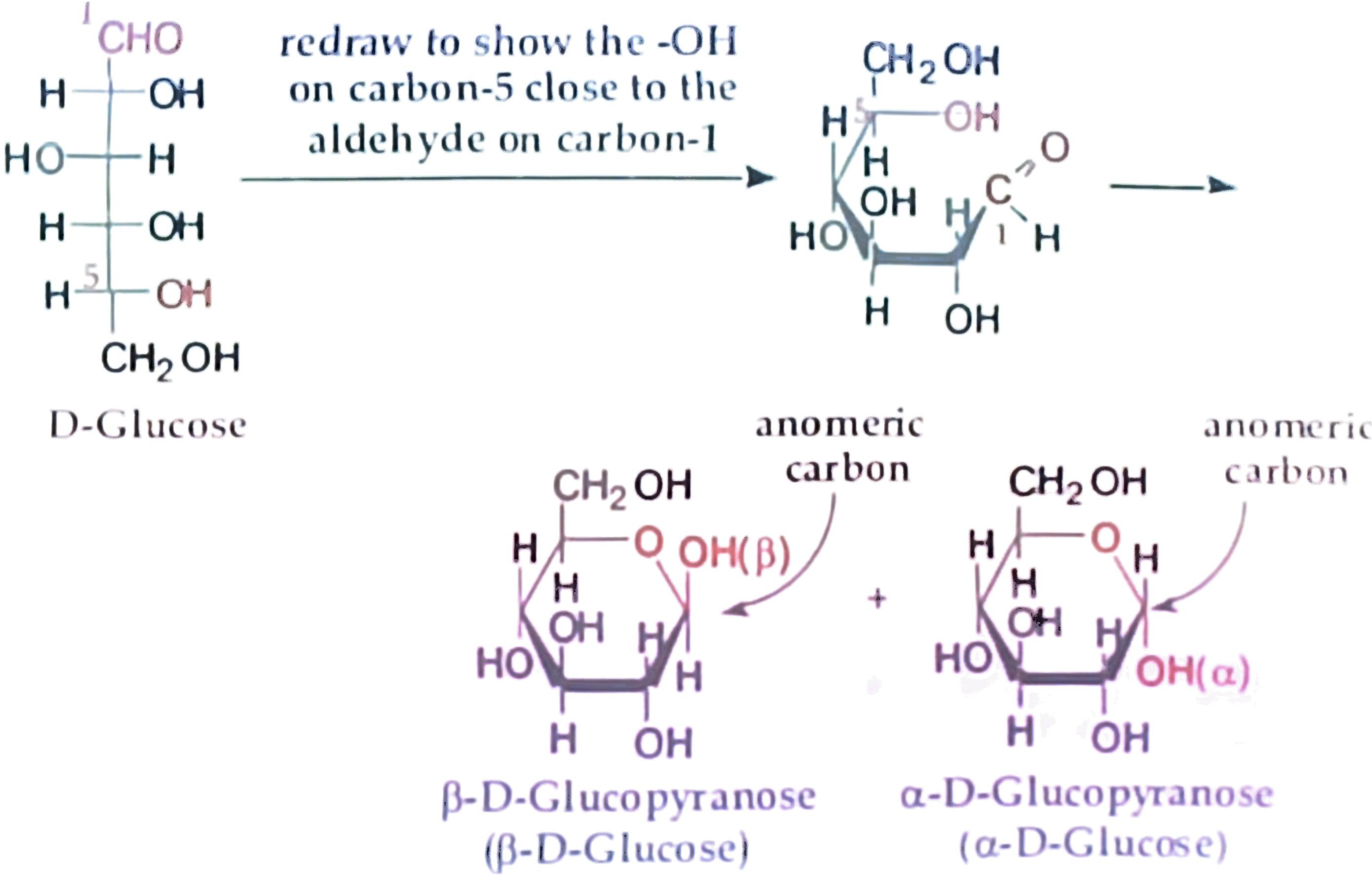
Haworth Projections

- six-membered hemiacetal rings are shown by the infix -pyran- so become pyranose.
- five-membered hemiacetal rings are shown by the infix -furan- so become furanose.

Furan Pyran

Hemiacetal is a functional group that forms when an alcohol reacts with an aldehyde or ketone




CHO
H OH
HO H
H OH
H OH
CH2OH
D-Glucose
redraw to show the -OH on carbon-5 close to the aldehyde on carbon-1
anomeric carbon
OH(β)
anomeric carbon
OH(α)
β-D-Glucopyranose
(β-D-Glucose)
α-D-Glucopyranose
(α-D-Glucose)

25.2 Cyclic Structure

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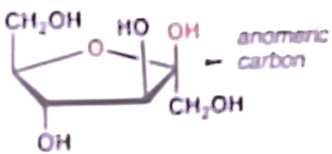

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25.2 Cyclic Structure

- *Monosaccharides have hydroxyl and carbonyl groups in the same molecule and exist almost entirely as five- and six-membered cyclic hemiacetals.*
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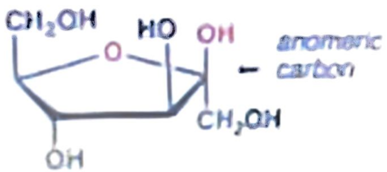
  - **anomers:** carbohydrates that differ in configuration at their anomeric carbons.



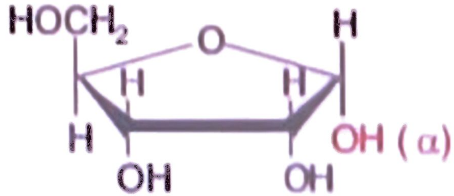
Anomer – the name given to two diastereomeric monosaccharides that are epimers at the anomeric carbon. This is C-1 in aldoses, and C-2 in the case of fructose

A. Haworth Projections

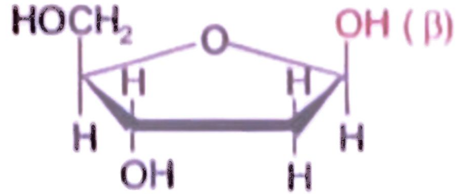
- Haworth projections:
  - five- and six-membered hemiacetals are represented as planar pentagons or hexagons, as the case may be, viewed through the edge.
  - they are most commonly written with the anomeric carbon on the right and the hemiacetal oxygen to the back right.
  - the designation β- means that the -OH on the anomeric carbon is *cis* to the terminal $-CH_2OH$; α- means that it is *trans* to the terminal $-CH_2OH$.

B. Conformational Formulas

- five-membered rings are so close to being planar that Haworth projections are adequate to represent furanoses.



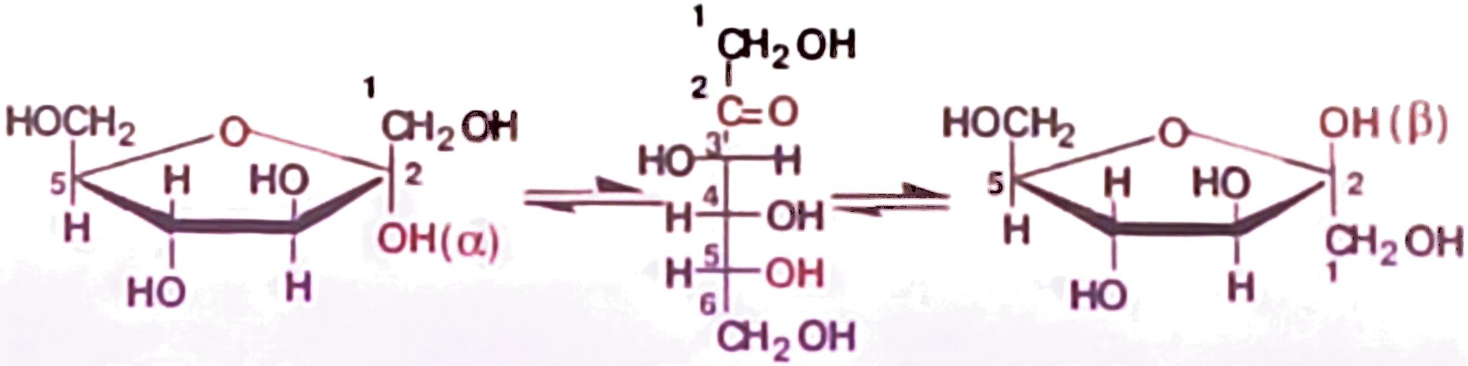
α-D-Ribofuranose
(α-D-Ribose)



β-2-Deoxy-D-ribofuranose
(β-2-Deoxy-D-ribose)

Conformational Formulas

- other monosaccharides also form five-membered cyclic hemiacetals.
- here are the five-membered cyclic hemiacetals of D-fructose.

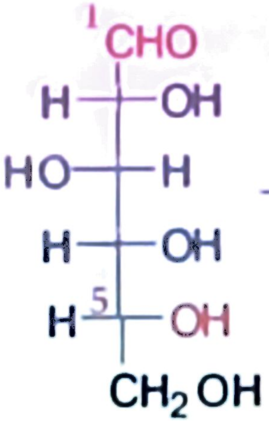


α-D-Fructofuranose
(α-D-Fructose)

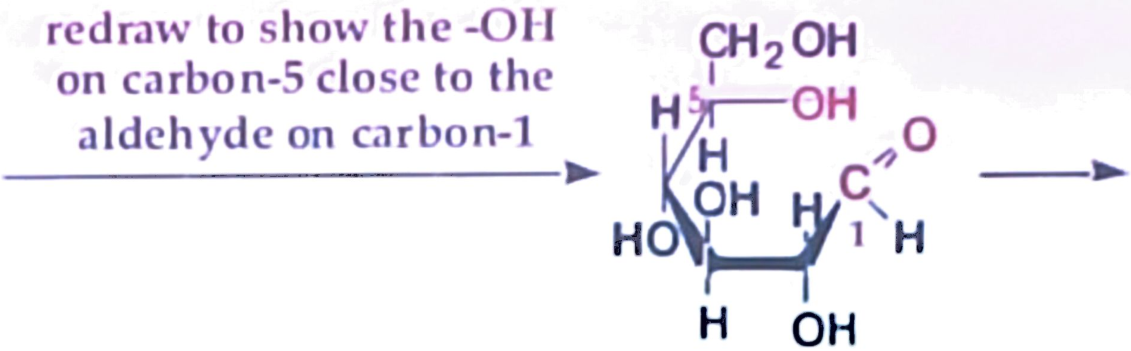
D-Fructose

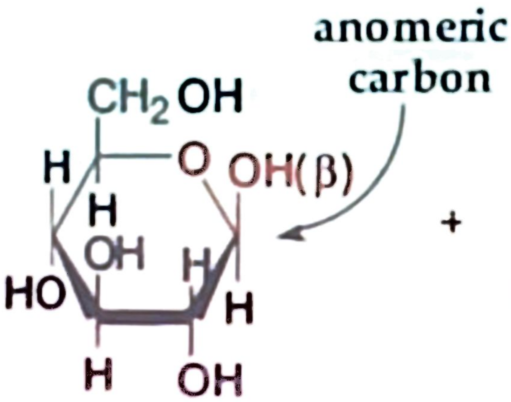
β-D-Fructofuranose
(β-D-Fructose)

Haworth Projections

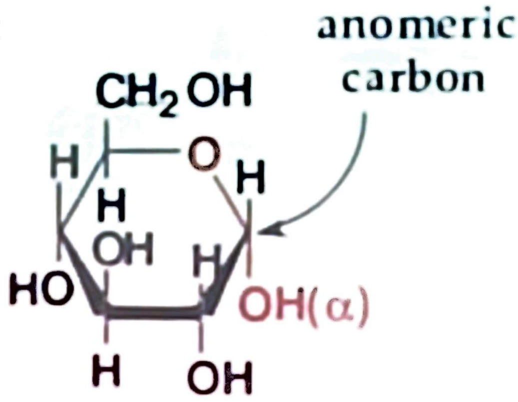


D-Glucose





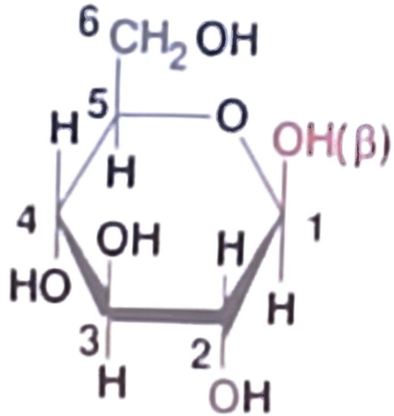
+



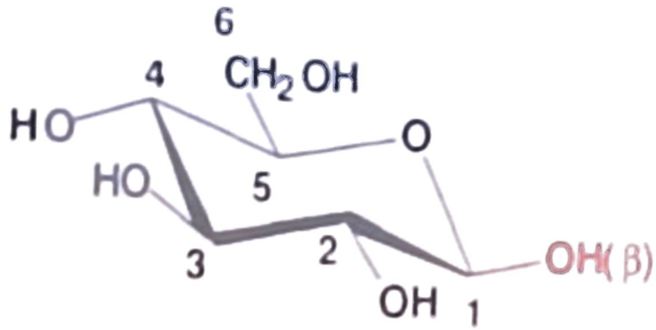
β-D-Glucopyranose
(β-D-Glucose)

α-D-Glucopyranose
(α-D-Glucose)

- if you compare the orientations of groups on carbons 1-5 in the Haworth and chair projections of β-D-glucopyranose, you will see that in each case they are up-down-up-down-up respectively.



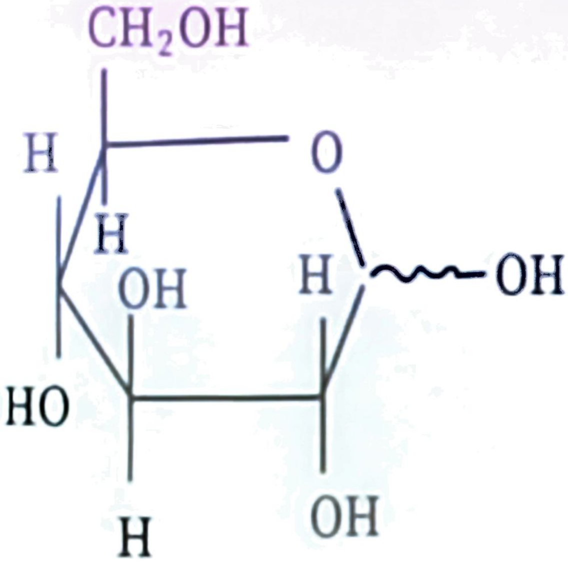
β-D-Glucopyranose
(Haworth projection)



β-D-Glucopyranose
(chair conformation)

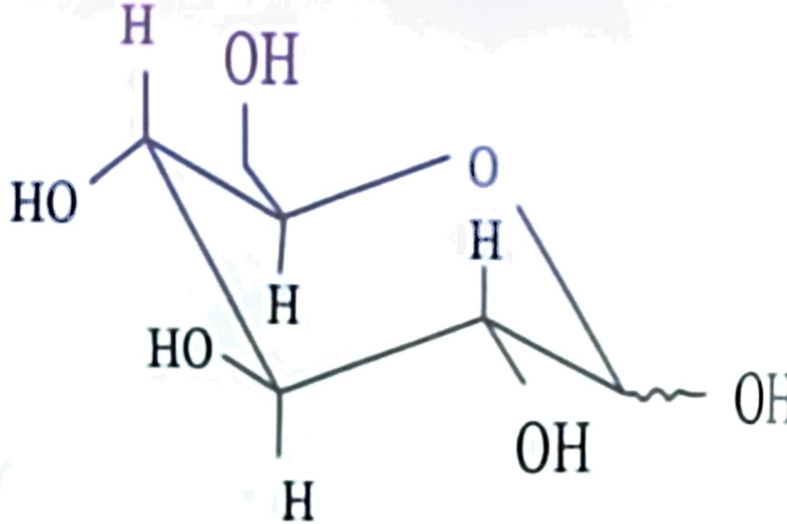
Conformational Formulas

- for pyranoses, the six-membered ring is more accurately represented as a chair conformation.



D-glucopyranose

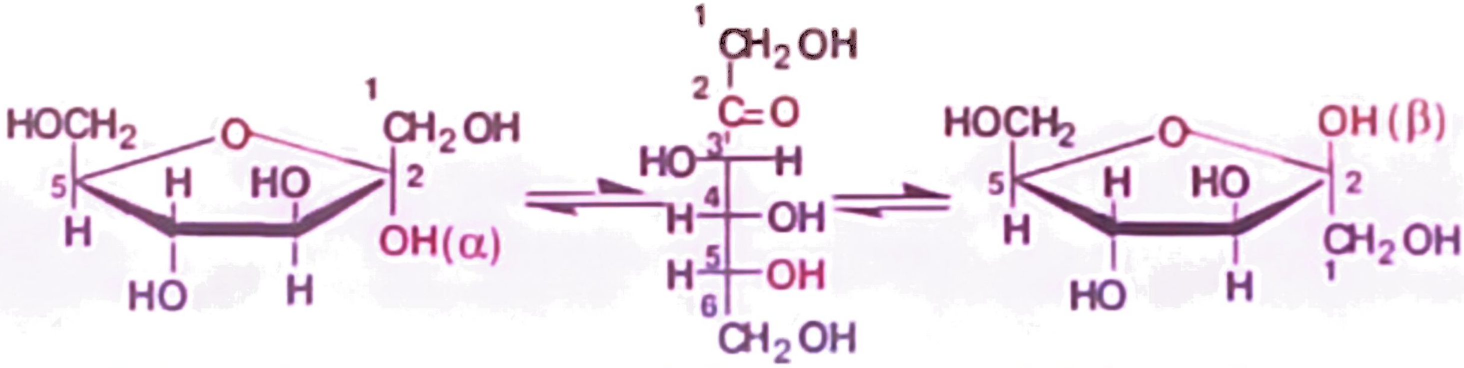
$\equiv$



chain form of D-glucopyranose

Conformational Formulas

- other monosaccharides also form five-membered cyclic hemiacetals.
- here are the five-membered cyclic hemiacetals of D-fructose.



α-D-Fructofuranose
(α-D-Fructose)

D-Fructose

β-D-Fructofuranose
(β-D-Fructose)

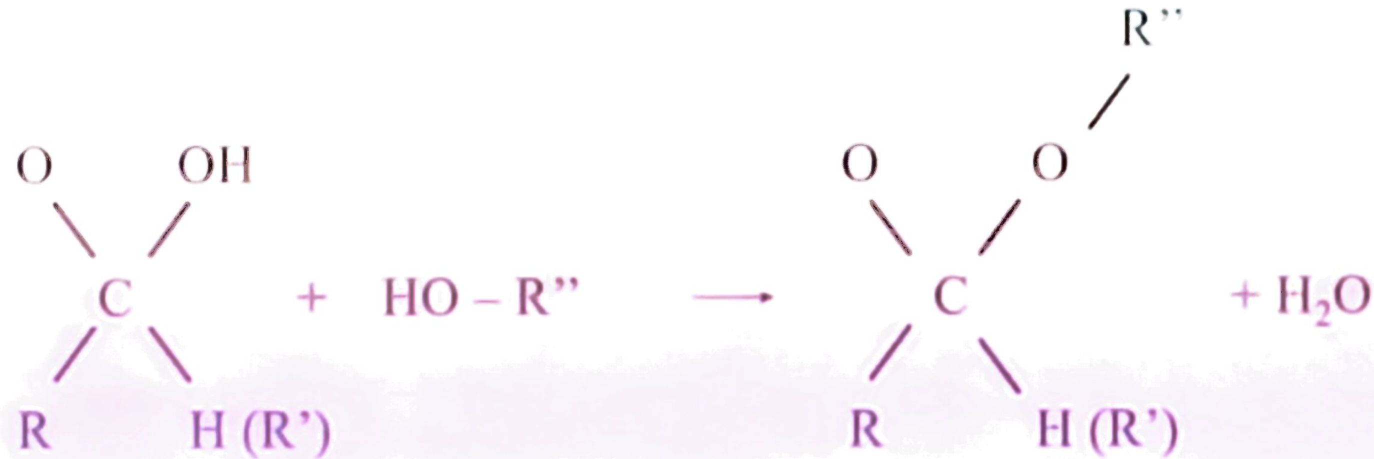
Glycosidic bonds

Bond formed between the anomeric carbon of a carbohydrate and the hydroxyl oxygen atom of an alcohol (O-glycosidic bond) or the nitrogen of an amine (N-glycosidic bond)

Glycosidic bonds between monosaccharides yields oligo- and polysaccharides

After glycosidic bond formation, the ring formation involving the anomeric carbon is stabilized with no potentially free aldehyde or keto groups

O-glycosidic bonds

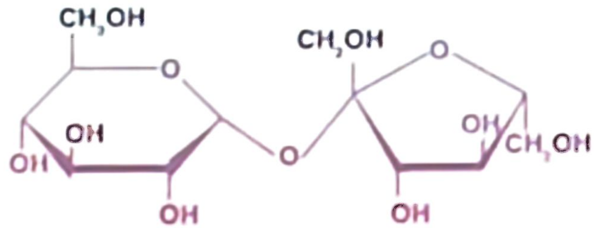


Hemiacetal (Hemiketal) | Alcohol | Acetal (Ketal) | Water

Disaccharide Examples

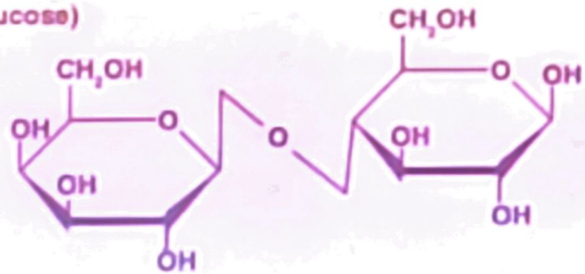
Sucrose

(Glucose-fructose)



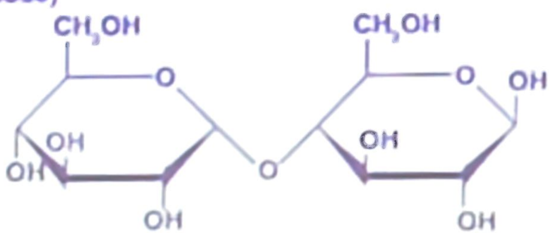
Lactose

(Galactose-glucose)



Maltose

(Glucose-glucose)



Sucrose
glucose + fructose

Lactose
galactose + glucose

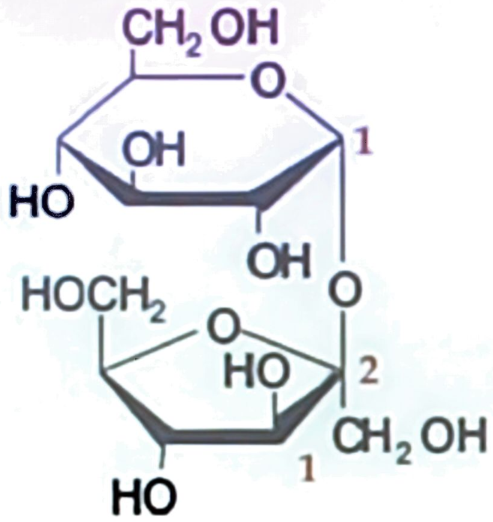
Maltose
glucose + glucose

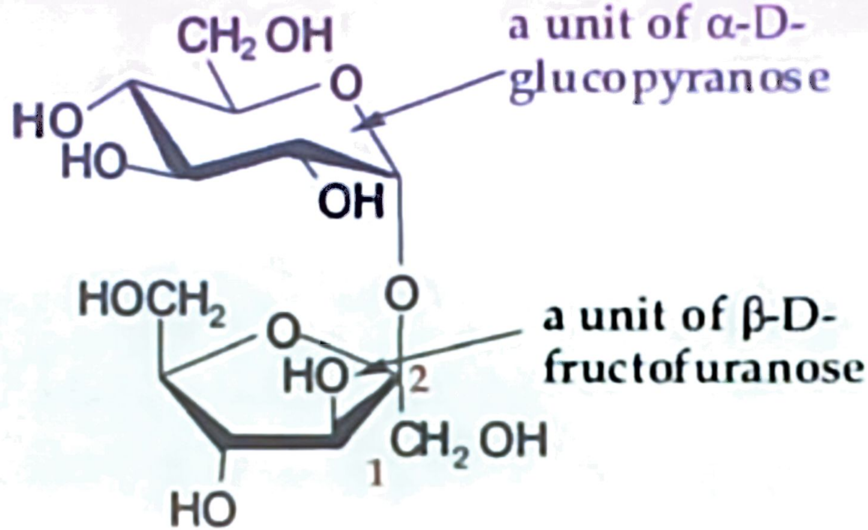
Sugar 1	+	Sugar 2	+	Linkage	=	Disaccharide
D-Glucose	+	D-Glucose	+	α-1,4 linkage	=	Maltose
D-Galactose	+	D-Glucose	+	β-1,4 linkage	=	Lactose
D-Glucose	+	D-Fructose	+	α-1,β-2 linkage	=	Sucrose

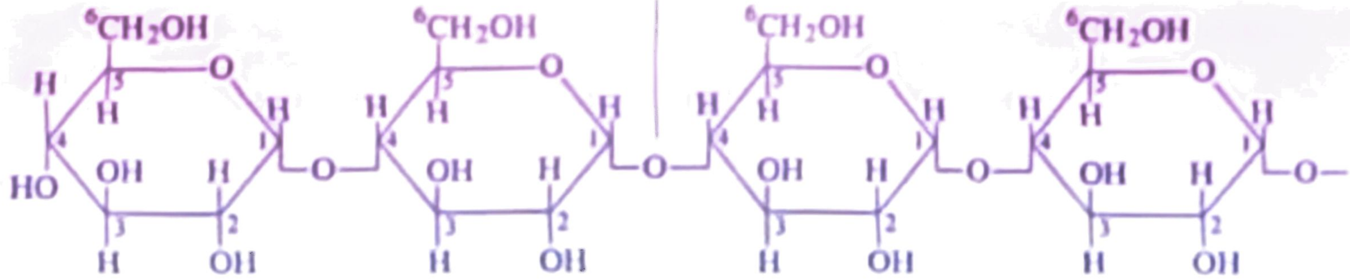
Upon hydrolysis, sucrose yields equimolar mixture of glucose and fructose which is often called invert sugar.

25.4 Disaccharides: A. Sucrose

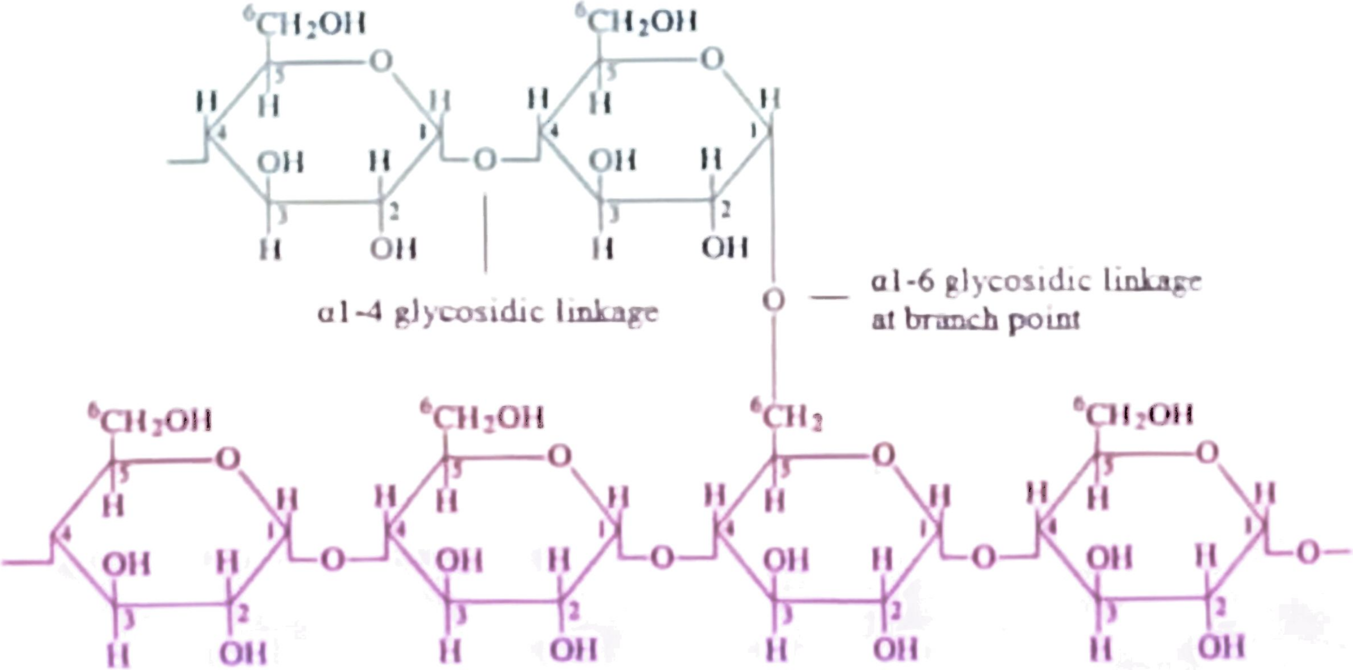
- Table sugar, obtained from the juice of sugar cane and sugar beet.







Amylose



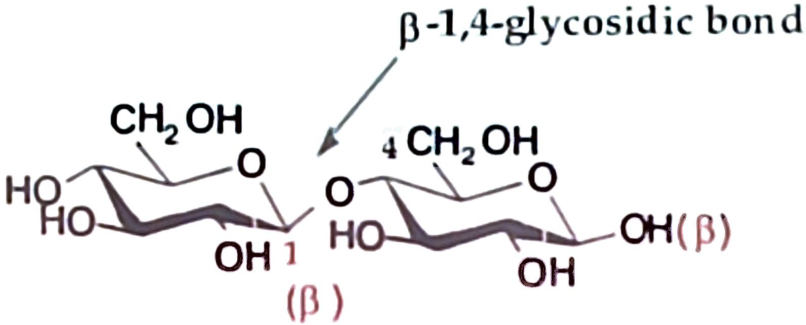
Amylopectin

25.5 Homopolysaccharides: A. Starch

- Starch is used for energy storage in plants.
  - it can be separated into two fractions; amylose and amylopectin; each on complete hydrolysis gives only D-glucose.
  - Amylose is composed of continuous, unbranched chains of up to 4000 D-glucose units joined by α-1,4-glycosidic bonds.
  - amylopectin is a highly branched polymer of D-glucose; chains consist of 24-30 units of D-glucose joined by α-1,4-glycosidic bonds and branches created by α-1,6-glycosidic bonds.

Disaccharides: D. Cellibiose

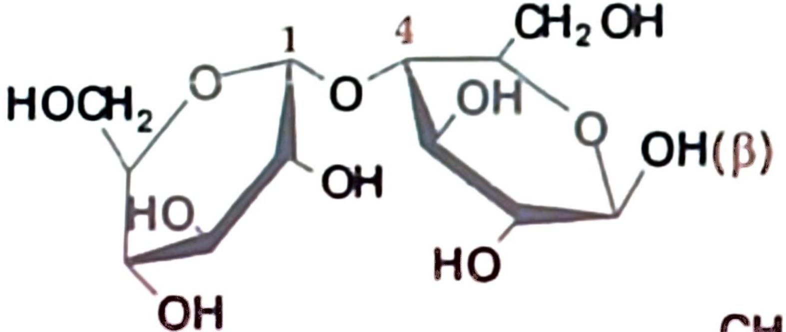
- Disaccharide from cellulose. Cellulose is the structural carbohydrate in plants.

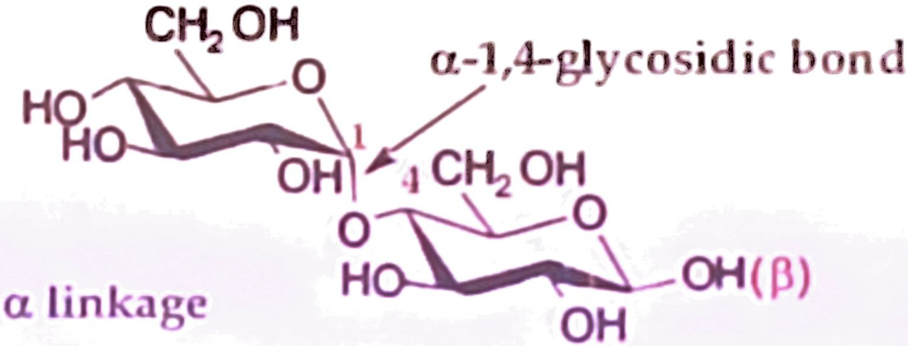


Glucose-glucose with a β linkage

Disaccharides: C. Maltose

- Disaccharide from starch or glycogen. From malt, the juice of sprouted barley and other cereal grains.

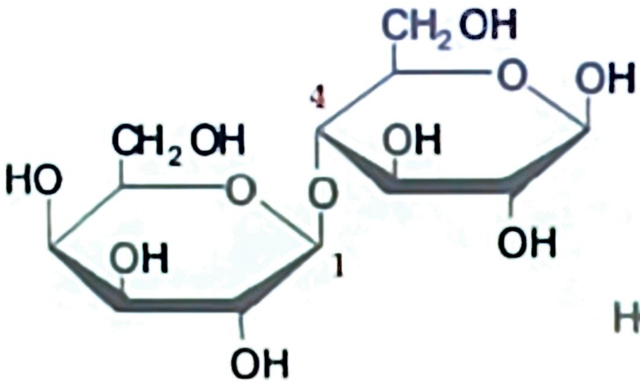


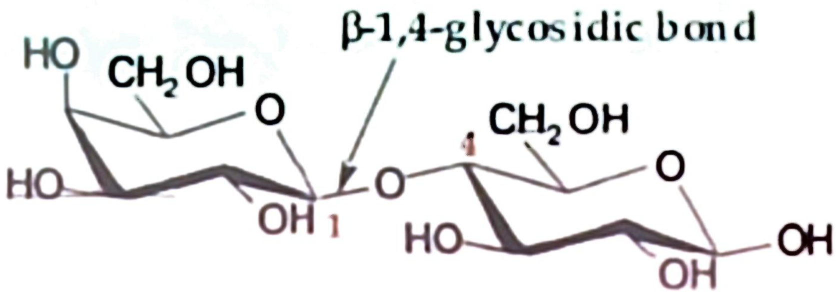


Glucose-glucose with an α linkage

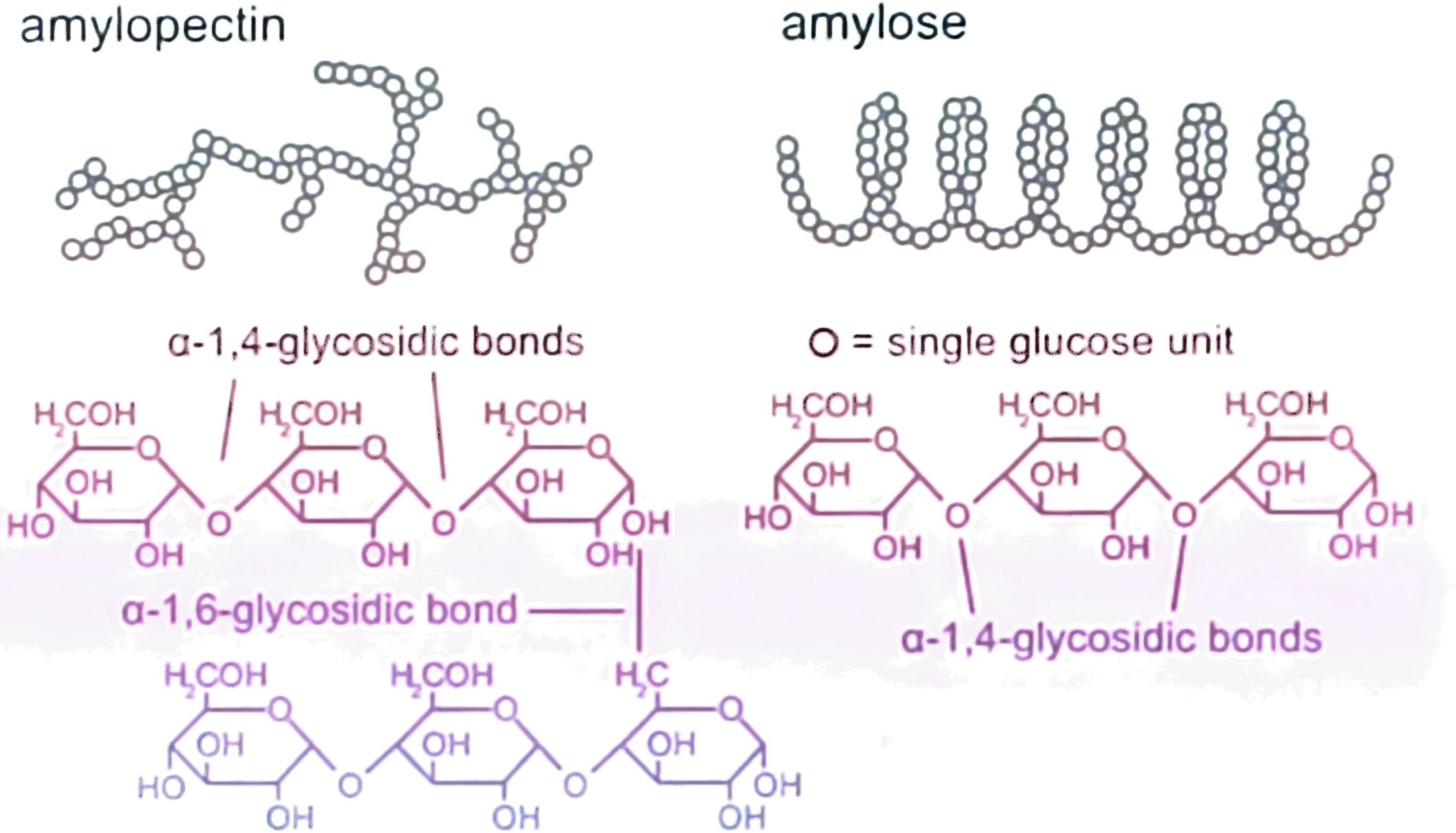
Disaccharides: B. Lactose

- The principle sugar present in milk.
  - about 5 - 8% in human milk, 4 - 5% in cow's milk.





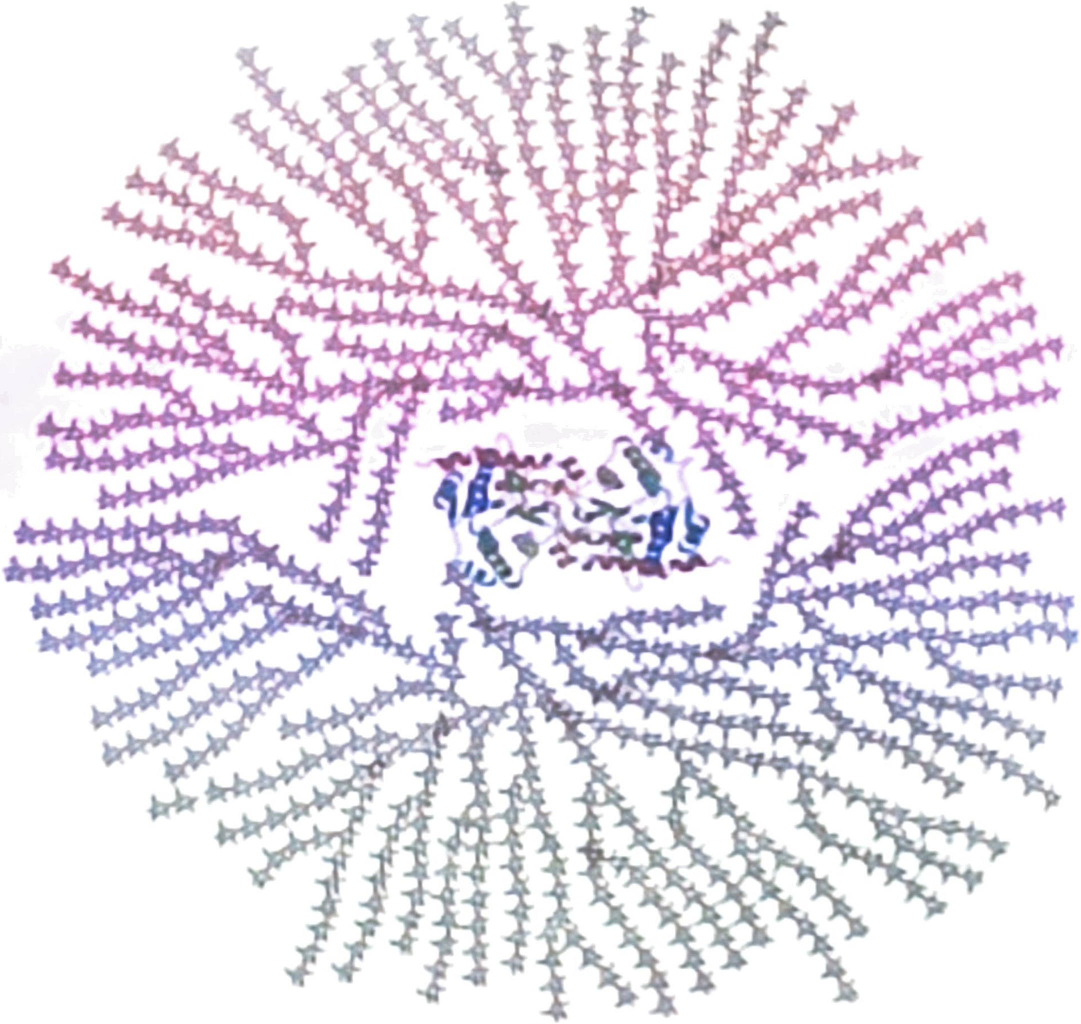
Galactose-glucose with a β linkage


amylopectin
amylose
α-1,4-glycosidic bonds
O = single glucose unit
α-1,6-glycosidic bond
α-1,4-glycosidic bonds

Homopolysaccharaides: B. Glycogen

- Glycogen is the reserve carbohydrate for animals.
  - like amylopectin, glycogen is a nonlinear polymer of D-glucose units joined by α-1,4- and α-1,6-glycosidic bonds bonds.
  - the total amount of glycogen in the body of a well-nourished adult is about 350 g (about 3/4 of a pound) divided almost equally between liver and muscle.

A core protein of glycogenin is surrounded by branches of glucose units. The entire globular granule may contain around 30,000 glucose units



Homopolysaccharaides: C. Cellulose

- Cellulose is a linear polymer of D-glucose units joined by β-1,4-glycosidic bonds.
  - it has an average molecular weight of 400,000 g/mol, corresponding to approximately 2800 D-glucose units per molecule.